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Multi-Objective Path Optimization of Highway-Railway Multimodal Transport Considering Carbon Emissions

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Abstract: With the increase in carbon emissions from railway transit, green transportation has attracted worldwide attention due to its low pollution and low consumption. In order to improve the transportation efficiency of multimodal transport and reduce carbon emissions, this paper makes a systematic study on the comprehensive optimization model and method of multiple transport tasks and transport modes considering carbon emissions. Firstly, an optimization model is established with transportation distance, transportation time, and carbon emission as transportation objectives. Secondly, an improved fuzzy adaptive genetic algorithm is designed to adaptively select crossover and mutation probabilities to optimize the path and transportation mode by using population variance. Finally, an example is designed, and the method proposed in this paper is compared with the ordinary genetic algorithm and adaptive genetic algorithm, which proves the proposed model and algorithm are effective. In conclusion, it is found that the present multi-objective optimization model based on the improved genetic algorithm can adjust multimodal transport plans and reduce carbon dioxide emissions, which provides a reference basis for logistics enterprises to carry out multimodal transport.

Keywords: multi-objective optimization; carbon emissions; FAGA algorithm; multimodal transport; path optimization

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1. Introduction

1.1. Research Status

In recent years, with the continuous improvement of infrastructure, the carbon emissions of the transportation industry have increased. According to the data from the International Energy Agency, the transportation industry is the second largest carbon emission field in the world, accounting for 25% of carbon emissions. China's transportation sector accounts for about 15% of the country's terminal emissions and is the third largest carbon emission source after industry and construction [1]. Facts have proved that it is difficult for a single mode of transportation to meet the requirements of the market, the economy, and the environment [2,3]. Multimodal transport uses a variety of transport services (railway, highway, air, and sea) to transport cargoes from their origin to their destination. Combining the different advantages of multiple transport services can significantly improve transport efficiency, reduce logistics costs, reduce total carbon emissions, enhance transport capacity, expand transport scope, and promote economic development along the route, which is in line with the concept of sustainable development [4–6]. In the critical period of the transformation and upgrading of the logistics industry, it is absolutely critical for us to improve the service quality of the whole process of transportation and reduce carbon emissions. The development of multimodal transport to achieve the effective integration of different resources is conducive to speeding up the construction of an integrated transportation system. The research on the path optimization of multimodal transport will effectively promote the transportation industry to develop in a more economical, reliable, flexible, and sustainable direction [7–9].

1.2. Focus of This Study

Based on the latest progress of highway-railway multimodal transport, this study studies the multimodal transport route selection problem from the perspective of improving the efficiency of transportation and making full use of the advantages of various modes of transportation. Firstly, a deterministic path planning problem model is established, which comprehensively considers transportation cost, transportation time, and carbon emissions. Secondly, the three are taken as the objective function of the model. Finally, a fuzzy adaptive genetic algorithm (FAGA) with adaptive selection of crossover probability and mutation probability is designed to solve the design case. From the perspective of multimodal transport operators, the single-objective model and multi-objective model of low-carbon multimodal transport path optimization are established, respectively.

(1) A three-objective function and a multi-constraint model of cost, time, and carbon emissions are established to comprehensively consider the factors considered by customers and multimodal carriers in the transportation process.

(2) The fuzzy inference system is model independent. It does not need the mathematical description of the functional relationship between input and output but learns from samples. At the same time, the fuzzy inference system has a parallel structure, is simple to operate, and requires only a small amount of computation. We compare FAGA with ordinary GA, adaptive GA, and elite GA in [10] to test its effectiveness. The results show that FAGA can select a better intermodal path and is significantly better than other improved genetic algorithms.

2. Literature Review

Research Status

The main direction of the existing multimodal transport route optimization research is to consider the quality of service and carbon emissions.

Considering the cost objective function, the literature used to focus on finding the optimal path considering quality from the perspective of multimodal transport operators, paying attention to the cost of the transportation process, and improving service quality and user satisfaction [11–14]. Milan [11] proposed a model for calculating the internal and external comprehensive costs of intermodal transport and highway freight transport networks. Considering other consumption in the process of transportation, Guan [12] comprehensively considered factors such as the transportation cost, transportation safety, transportation route, and transportation mode and established a multimodal transportation route selection model with the lowest transportation cost as the objective function. Wang [13] considered a discrete multimodal transport network design problem, established or extended a path to minimize the total operating cost, and summarized the multi-cost problem into a total operating cost, which greatly reduces the computational complexity. Liu [14] considered the transportation and purchase costs of soybeans under the background of the global containerized transportation of soybeans, and a multimodal transport network model was established to optimize the transportation route of soybeans. Most of the above studies only considered the factor of transportation cost, but the evaluation of the route in the operation plan often considered many factors such as transportation time and transportation cost, so it lacked comprehensiveness. Ma [15] constructed an opportunity-constrained programming model for a class of vehicle routing problems considering customer classification, random travel time, random service time, and time window constraints. In addition to pursuing the minimization of transportation time and cost from the perspective of the shipper and from the perspective of the customer, Hao [16] used the dynamic programming method to solve the transportation path. Gräbener [17] considered all the consumption time in the transportation process to establish a time-related multi-objective optimal path optimization model and gave a variety of adaptive decision preference suggestions, which were more comprehensive. Fan [18] took into account the vehicle routing problem of simultaneous pickup under the condition of time-varying road speed and soft time windows. The path optimization model was established with the goal

of minimizing the sum of vehicle dispatching cost, time window penalty cost, and vehicle transportation cost. It added more constraints and emergencies that may be encountered during transportation.

Although most of the above studies have conducted research on route integration and considered many factors in route optimization, most of them only considered a single mode of transportation and did not consider the comprehensive optimization of transport routes from the perspective of multimodal transport in combination with the current situation.

The existing literature on multimodal transport path optimization mainly focuses on the model and algorithm of a single-task problem with time window constraints [19–21]. However, the problem of carbon emissions in the transportation process is generally ignored, and most existing studies do not consider the multimodal transport path optimization under multi-task conditions. With the development of science and technology and the demand for green sustainable development, the goal of multimodal transport has changed from considering costs to paying attention to the impact of transportation on the environment.

With the development of smart cities, new requirements have been put forward for the control of carbon emissions (CEs) in the transportation system [22–24]. Chen [25] combined low-carbon issues and logistics issues in the process of research long ago and analyzed the significance of logistics operation mode decision-making on carbon emission reduction through relevant mathematical models, which has a certain inspiration for similar research. Considering the randomness and uncertainty of energy consumption, Li [26] used stochastic optimization theory to establish a green multimodal transport path optimization model under mixed uncertainty with transportation costs, carbon emission costs, and time penalty costs as objectives. From the perspective of multimodal transport operators, Cheng [27] established a multi-task multimodal transport path comprehensive optimization model considering carbon emissions. Although the above research takes into account the impact of carbon emissions in transportation problems such as logistics operations, the research on the introduction of multimodal transport route optimization in combination with the exchange time cost of intermodal goods is still blank. The literature [10,28–30] mainly studied the comprehensive optimization models and methods of routes and modes of transportation for multiple transportation tasks considering carbon emissions. Lam [31] combined the factors of cost, time, and energy consumption to establish a multimodal transport network optimization model that meets the requirements of time, cost, and carbon emissions. Based on the research background of China–EU container intermodal transportation, Li [32] highlighted the optimization objectives of time, cost, and carbon emissions, constructed the optimization model of a China–EU container intermodal transportation path, and designed a genetic algorithm to solve it. Jiang [33] focused on the issue of carbon emissions. Since the gas that produces carbon emissions is mainly carbon dioxide in multimodal transport, he studied the multimodal transport network with carbon dioxide emission reduction targets and pollution emissions. Qu [34] linearized the nonlinear integer programming model and analyzed the cost changes of single transport modes and multimodal transport schemes based on considering carbon emissions.

On the basis of the model objectives and carbon emission constraints, it is necessary to further solve the multimodal transport path optimization problem. Therefore, in recent years, most studies have begun to focus on multi-objective optimization algorithms. The existing research has often used the weight coefficient method for multi-objective models, which expresses the relative importance of the factors with a certain weight and carries on the weighted summation to realize the transformation from multi-objective to single-objective tasks [35–37]. The research of a multimodal transport route optimization algorithm involves the Dijkstra algorithm, genetic algorithm, improved genetic algorithm, ant colony algorithm, the combination of genetic algorithm and Dijkstra algorithm, genetic algorithm based on k shortest path, and so on. Lam and Gu [31] established an optimization model of a multimodal transport network aiming at minimizing cost, time, and carbon emissions, designed a genetic algorithm to solve it, and analyzed an example of import and

export containers in an inland city in China. They proposed the influence of coordinated transition time and different carbon emission limits on the design of their multimodal transport network. However, its essence was a dual-objective optimization model. Based on the traditional genetic algorithm, the constraints of the optimization problem could not be fully expressed, which increased the workload and solution time, and the goal of the lowest carbon emissions was not considered. Secondly, the literature [31] used the general MILP solver cplex12.4 to obtain the Pareto optimal set. It did not use some meta-heuristic algorithms needed to solve large-scale cases but used the solver to solve small-scale calculation problems. The solution method has limitations. The literature [38] introduces a singleton Type 1 fuzzy logic system (T1-SFLS) controller and fuzzy-WDO hybrid for autonomous mobile robot navigation and collision avoidance in an unknown static and dynamic environment. The computational performance of the algorithm was optimized by considering fuzzy factors. The literature [39] proposed a path-planning method of collision avoidance for multi-ship encounters that is easy to realize for autonomous ships. Aiming at producing a high-quality flight path for unmanned autonomous helicopters with multi-constraints, a path planning method was proposed based on the multi-strategy evolutionary learning artificial bee colony algorithm in [40]. An artificial bee colony algorithm is a kind of swarm intelligence algorithm, which uses less control parameters and increases robustness.

These studies have fuzzified some difficult-to-quantify parameters to be considered in algorithms and path optimization objects, fuzzified the unmodeled dynamics of the transportation process into the model, improved the effectiveness of optimization, and given full play to the potential of fuzzy systems to solve path optimization. The existing research results also foreshadow the possibility of using fuzzy for algorithmic optimization. The research content of the relevant literature is shown in Table 1.

Aiming at the problems of complexity, low efficiency, and serious environmental pollution in container multimodal transportation, this paper introduces the fuzzy reference system, uses the improved genetic algorithm to optimize the path, and establishes a multimodal transportation comprehensive optimization model and solution algorithm considering the three objectives of cost, time, and carbon emission. It provides a reference for logistics enterprises to carry out multimodal transport by guiding multimodal operators to adjust transport schemes and reduce carbon emissions.

Table 1. Research status analysis.

Literature	Transportation Ways	Algorithm Type	Objective Function				Constraint Condition			
			Transportation Cost	Transferring Cost	Carbon Emissions	Other	Transportation Time	Freight Volume	Carbon Emission	Traffic Capacity
[11]	Highway, railway, waterway	Heuristic algorithm	✓							
[12]	Highway, railway, waterway	Heuristic algorithm	✓				✓			
[13]	Highway, railway, waterway	Heuristic algorithm	✓							
[14]	Highway, railway, waterway	Heuristic algorithm	✓							
[15]	Highway	Heuristic algorithm	✓	✓		Damage cost	✓	✓		✓
[16]	Highway, railway, waterway	Heuristic algorithm	✓	✓	✓	Time penalty cost	✓	✓		✓
[17]	Bicycles, walking, public transport	Greedy algorithm	✓	✓	✓	Time penalty cost	✓	✓		✓
[18]	Highway	Genetic algorithm	✓	✓		Time	✓	✓		
[19]	Highway, railway, waterway	Genetic algorithm	✓		✓					
[20]	Highway, railway, waterway	Genetic algorithm	✓			Time and mileage				✓
[26]	Highway, railway, waterway	Heuristic algorithm	✓		✓	Time penalty cost				
[27]	Highway, railway, waterway	Greedy algorithm			✓					
[28]	Highway	Simulated annealing arithmetic			✓					
[29]	Highway	Ant algorithm			✓		✓			✓
[10]	Highway, railway, waterway	Genetic algorithm	✓	✓		Damage cost				
[31]	Highway, railway, waterway	Heuristic algorithm	✓		✓	Time				
[32]	Highway, railway, waterway	Genetic algorithm	✓		✓	Time				
Method of this paper	Highway, railway, waterway	Fuzzy adaptive genetic algorithm			✓	Time and mileage				

3. Establishment of the Model

3.1. Problem Description

The multimodal transport route optimization strategy considering carbon emissions is a complex problem that integrates multiple modes of transportation, multiple transport routes, multiple transit nodes, fixed starting and ending points, and multiple random factors and fully takes into account the reduction of input costs of logistics enterprises, improvement of transport efficiency, reduction of carbon emissions, and other factors. In the whole route planning, this paper will select three factors: time, cost and carbon emissions, to build a multimodal transport route optimization scheme.

It is assumed that a batch of goods needs to be transported from city A to city G in containers, and there are several nodes. The random path layout network diagram of multi-modal transport is shown in Figure 1. As can be seen from the layout network diagram of Figure 1, from city A to city G, there are five nodes: B, C, D, E, and F. Two different modes of transportation can be selected between each node. According to the carbon emissions between the nodes, the possible transportation time and cost of the two modes of transportation are compared, and finally, the optimal combination of different modes of transport is found to complete the multi-modal transport scheme from A to G.

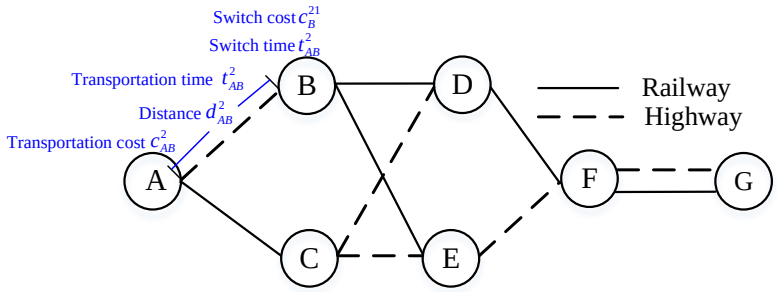


Figure 1. Network diagram of random path layout for multimodal transport.

The problems that need to be solved in the research on the route optimization of public-railway multimodal transport considering carbon emissions can be described as follows: on the multimodal transport network, find a train route covering all the intermediate nodes of the network. Take transport cost, transport time, and carbon emissions as objective functions, carbon emissions, transport time windows, and transport modes as constraints, calculate the train service frequency, and meet the needs of freight transport with the minimum operating cost.

3.2. Model Hypothesis

It is assumed that in the process of transporting goods from the starting city to the destination city, there are highway transport and railway transport. The carbon dioxide emissions generated by different transport modes are different because of their different transport time, distance, and cost. At this time, multimodal transport operators need to meet the time needs of customers, combined with the carbon emission limits of logistics enterprises. An optimal transportation path from the starting point to the destination is obtained. Transportation is the key to logistics cost management and savings, and the cost value of multimodal transportation is a key reference indicator for operators to adjust transportation solutions. Transportation time is the key indicator in multimodal transport issues. Goods arrive early or late and need to pay storage or delay costs. Goods that arrive at the end time must meet the end time window constraints; otherwise, user satisfaction will be affected. With the increasing socio-economic growth, low-carbon green development also puts forward new requirements for the transportation industry, which means that achieving low-carbon transportation in multimodal transport is of great significance to the improvement of transportation quality. On this basis, this paper selects transportation cost, transportation time, and carbon emissions as the optimization objectives. To describe the

transportation problem more appropriately and to facilitate the construction and solution of the model in this paper, the following assumptions are made for the problem.

(1) The capacity limitations of transport routes and transfer nodes are not considered;

(2) In the process of transportation, the freight volume remains fixed regardless of the increase or decrease of goods;

(3) The mode of transport between the two adjacent nodes is inseparable; that is, only one mode of transport can be chosen;

(4) The transfer of goods only occurs at the node, and the transit process occurs at most once at a node.

(5) Damage to goods, road congestion, and natural disasters are not taken into account in transportation.

The symbols of the parameters and decision variables involved in this question are as follows (Table 2):

Table 2. Symbol description of parameters and decision variables.

n	The set of transport nodes: A, B, C, ... G
h	The set of transport nodes: B, C, ... F
J	The set of transport modes: highway, railway
i	The ith node
j	The jth node generally refers to the next node of node i
k	The kth mode of transport
l	The lth mode of transport
q	Freight volume (unit: t)
S_i	The time when the goods arrive at the node i
$T_{\min}$	The lower limit of time allowed for the goods from the beginning of transportation to the end of transport
$T_{\max}$	The maximum amount of time allowed for the goods from the beginning of transportation to the end of the transport
d_{ij}^k	The distance between node i and node j using the transport mode k
c_{ij}^k	The unit transportation cost of the transport mode k between node i and node j
c_i^{kl}	The transportation cost of the node i from transport mode k to transport mode l
t_{ij}^k	The transportation time of the transport mode k between node i and node j
t_i^{kl}	Unit transport time of node i converted from mode k to mode l
e_k	Unit carbon emissions using the transport mode k
$E_{\max}$	Maximum allowable carbon emissions during transportation
x_{ij}^k	When the transport mode k is adopted between node i and node j, the value is 1 and 0 otherwise
y_i^{kl}	Node i takes a value of 1 when it is converted from transport mode k to transport mode l; otherwise, it is 0

3.3. Objective Function

3.3.1. Transportation Cost

(a) In-transit transport cost is related to the freight volume, unit transport cost, and transport distance loaded by the means of transport:

$$C_1 = \sum_{i \in n} \sum_{j \in n} \sum_{k \in J} q d_{ij}^k c_{ij}^k x_{ij}^k \quad (1)$$

(b) Regarding node conversion cost, the conversion cost will occur when a node changes the means of transport, which is related to the freight volume and the unit conversion cost between different modes of transport, as follows:

$$C_2 = \sum_{i\in h}\sum_{k\in J}\sum_{l\in J} q c_i^{kl} y_i^{kl} \tag{2}$$

3.3.2. Transportation Time

In this paper, transportation time is analyzed as an objective function. The capacity limits of different transportation modes between nodes and the transfer capacity limits of transit nodes are set. Transportation time is divided into transit time and node switching time.

(a) Transit time:

$$T_1 = \sum_{i\in n}\sum_{j\in n}\sum_{k\in J} t_{ij}^k x_{ij}^k \tag{3}$$

(b) Node switching time:

$$T_2 = \sum_{i\in h}\sum_{k\in J}\sum_{l\in J} q t_i^{kl} y_i^{kl} \tag{4}$$

3.3.3. Carbon Emission

This paper only studies the carbon emissions generated in the process of multimodal transport. The carbon emissions caused by the energy consumption of different means of transport are different, and the transport distance and the weight of goods also affect the carbon emissions. Therefore, by combining these influencing factors, the carbon emissions generated by the transportation process, in general, are as shown in Formula (5):

$$E = \sum_{i\in n}\sum_{j\in n}\sum_{k\in J} q d_{ij}^k e_k x_{ij}^k \tag{5}$$

3.4. Constraint Condition

$$s.t. \sum x_{i,i+1}^k = 1 \quad \forall i \in n \tag{6}$$

$$s.t. \sum y_i^{kL} = 1 \quad or \quad 0 \quad \forall i \in n \tag{7}$$

$$x_{i-1}^k + x_{i,i+1}^l \geq 2y_i^{kl} \quad \forall i \in n, k \in J, l \in J \tag{8}$$

$$E \geq E_{\max} \tag{9}$$

$$S_j = S_i + \sum_{j\in J}\sum_{z\in J} x_{ij}^k y_i^{kl} (t_i^{kl} + t_{ij}^k) \quad \forall_{i,j} \in I \tag{10}$$

$$T_{\min} \leq \sum_{i\in n}\sum_{j\in n}\sum_{k\in J} t_{ij}^k x_{ij}^k + \sum_{i\in n}\sum_{k\in J}\sum_{l\in J} y_i^{kl} t_i^{kl} \leq \mathrm{T}_{\max} \quad \forall_{i,j} \in n \tag{11}$$

$$x_{ij}^j \in \{0, 1\} \tag{12}$$

$$y_i^{kl} \in \{0, 1\} \tag{13}$$

Formula (6) means that only one mode of transportation can be chosen between two adjacent nodes; Formula (7) means that each city has only one chance to change; Formula (8)

indicates the logical relationship between variables, which ensure the continuity of cargo transportation; Formula (9) means that the carbon emissions of goods transportation cannot exceed the maximum allowable carbon emissions (this calculation is carried out according to the different energy consumption carbon emissions of railways and highways); Formula (10) uses a recursive method to express the time (this calculation is carried out according to the different transportation speeds of railways and highways) when the goods are transported to each location (node *i*); Formula (11) indicates that the goods are transported from the starting point to the endpoint, and the total transportation time and replacement time should be within the specified time window of the goods arriving at the destination; and Formulas (12) and (13) are the values of decision variables.

4. Design of Fuzzy Adaptive Genetic Algorithm

As the above construction is a multi-objective multimodal transport path optimization model, there are many variables involved in the model, and the network layout is also very complex, so it is more appropriate to use a genetic algorithm to solve the problem. The path optimization model studied in this paper is fundamentally a multi-objective combinatorial optimization problem, which belongs to an NP-hard problem and has the case of combinatorial explosion, so when solving this kind of problem, we generally use a heuristic search algorithm to reduce the search space and find a better solution as soon as possible. However, the optimization mechanism of multi-objective parallel optimization algorithms such as non-dominated genetic algorithm (NSGA-II) is to optimize multiple objectives at the same time, and the three objectives are of the same importance. In this case, the multimodal transport process should first consider that the cost demand is the key index for the operator to choose the path, and then consider the transportation time and transportation distance. The importance of the three objective models is different, so it needs to be multiplied by different weights and converted into a single objective solution. The solution is presented by a Pareto front containing many solutions, but compared to other traditional heuristic search algorithms, the genetic algorithm based on Pareto fitness shows better performance.

4.1. Algorithm Principle

In this paper, an improved fuzzy adaptive genetic algorithm (FAGA) is proposed. The genetic algorithm based on fuzzy logic makes full use of the existing knowledge and experience of GA and revises and perfects the experience. It is helpful to understand the relationship between GA genetic operators and parameter setting and GA performance. At the same time, in the process of GA operation, the dynamic adjustment of GA parameters or operators is realized, which ensures the reasonable utilization and exploratory relationship EER in the whole GA search process. By writing the rules, changing P_c and P_m into the fuzzy system, and inserting them into the genetic algorithm as a mechanism to adjust the two operators, we can adjust the parameters online in real time and optimize the performance of the algorithm. The algorithm uses the variance of population and individual fitness values to measure the overall population diversity and individual differences, applies it to design a fuzzy reference system that adaptively adjusts P_c and P_m, and automatically adjusts the parameters of the genetic algorithm. Fuzzy logic can effectively solve nonlinear problems with language and has fully proved its wide potential in the application of industrial automation. The basic principle of the FAGA algorithm is shown in Figure 2.

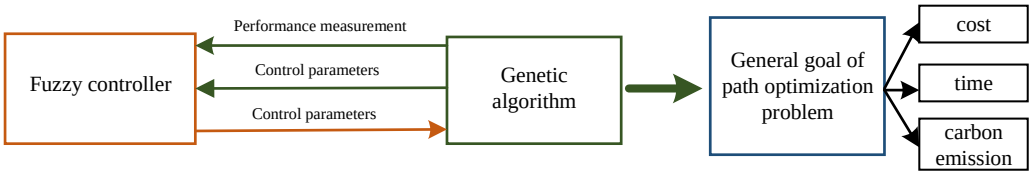


Figure 2. FAGA basic schematic diagram.

The fuzzy controller obtains the performance measurement and the parameters of P_c and P_m from the genetic algorithm and outputs the adjusted parameters to the optimization algorithm through fuzzy rule processing to achieve the purpose of an adaptive online adjustment algorithm. The optimization simulation results show that, compared with the standard genetic algorithm, the improved algorithm has faster convergence speed, better solution quality, and higher efficiency. The flow chart of the FAGA algorithm is shown in Figure 3.

The quantity that reflects the individual density in the population generally refers to the variance of the population adaptation value. In the process of evolution, the greater the variance, the more dispersed the spatial distribution of the population, and the lower the crossover probability and mutation probability, so that the population is rapidly concentrated; the smaller the variance, the smaller the mutation probability, and the crossover probability can be increased to avoid prematurity and improve diversity. Therefore, the variance of population fitness can be used to comprehensively express the concentration of the population. At the same time, the parameters of different individuals in each generation should be adjusted differently in each generation.

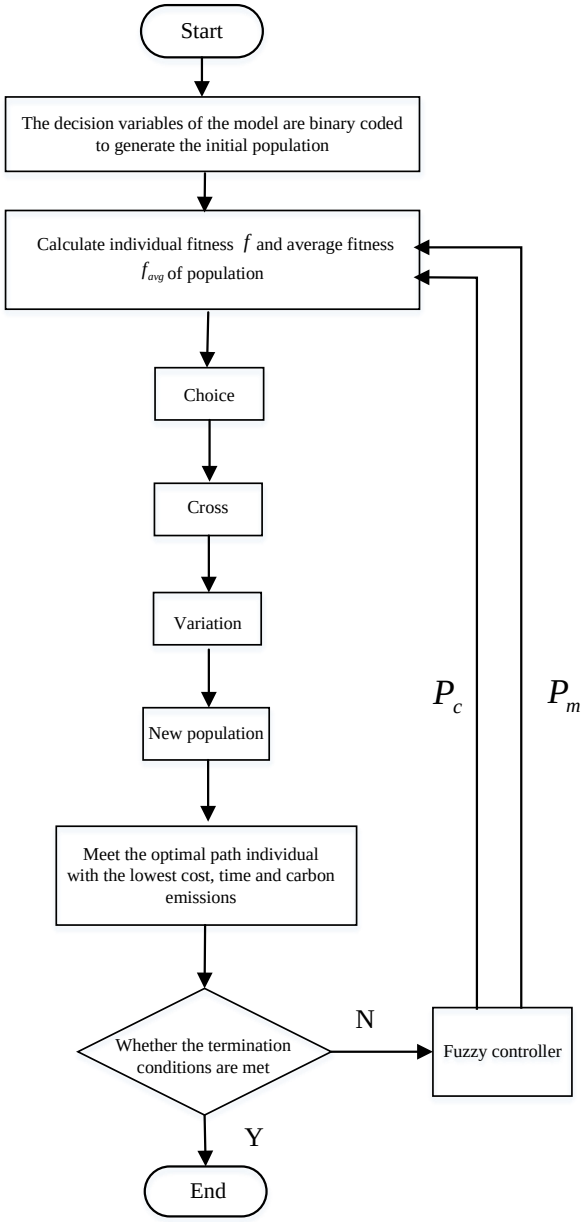


Figure 3. FAGA Basic flow chart.

In this paper, an adaptive genetic algorithm based on fuzzy rules is proposed, in which the fuzzy inference system based on Mamdani changes P_c and P_m to improve the population diversity of the algorithm. The calculation results show that this method has better performance than the general genetic algorithm. This is because it always adjusts the genetic operator to find the global optimal solution without reducing the population diversity, and the quality of the optimized solution is such that it is able to avoid falling into the local optimization.

The specific operation is as follows: the density of the population is measured by the variance of the distance between all individuals in the population and normalized as the input of the fuzzy system, the fuzzy language is transformed into the corresponding fuzzy variables of P_c and P_m , and the P_c and P_m in the algorithm are modified in defuzzification.

4.2. Algorithm Step

First of all, the three goals of the pre-population are averaged, and the three target values of cost, time, and carbon emissions, $\overline{x}_i$, $\overline{y}_i$, and $\overline{z}_i$, are calculated as shown in Formula (14).

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}, \bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}, \bar{z} = \frac{\sum_{i=1}^{n} z_i}{n} \tag{14}$$

where $\overline{x}_i$, $\overline{y}_i$, and $\overline{z}_i$ are the cost, time, and carbon emissions of each individual, and n is the number of the population. The distance difference between each individual and the intermediate individual is then calculated, as shown in Formula (15).

$$d_i = \sqrt{(x_i - \bar{x})^2 + (y_i - \bar{y})^2 + (z_i - \bar{z})^2} \tag{15}$$

where d_i is the distance between each individual and the intermediate value.

The mean square deviation V_d of the distance difference between all individuals and intermediate individuals is shown in Formula (16).

$$V_d = \sqrt{\frac{\sum_{i=1}^{n} d_i^2}{n}} \tag{16}$$

Finally, the input is normalized as follows, as shown in Formula (17).

$$F = \frac{V_d - V_{d\min}}{V_{d\max} - V_{d\min}} \tag{17}$$

where $V_{d\max}$ is the mean square deviation of the distance of the largest individual in the population, and $V_{d\min}$ is the mean square deviation of the distance of the smallest individual in the population.

For simplicity, the membership function of a fuzzy system is trapezoidal:

$$f(x) = \begin{cases} 0 \quad \text{if} \quad x \le a \\ \frac{x-a}{b-a} \quad \text{if} \quad a < x \le b \\ 1 \quad \text{if} \quad \text{b} < x \le c \\ \frac{d-x}{d-c} \quad \text{if} \quad c \le x \le d \\ 0 \quad \text{if} \quad x > d \end{cases} \tag{18}$$

where parameters a and d determine the bottom edge of the trapezoid, while parameters b and c determine the upper edge of the trapezoid.

The membership functions of mean square deviation F, crossover probability P_c, and mutation probability P_m are shown in Figure 4 below.

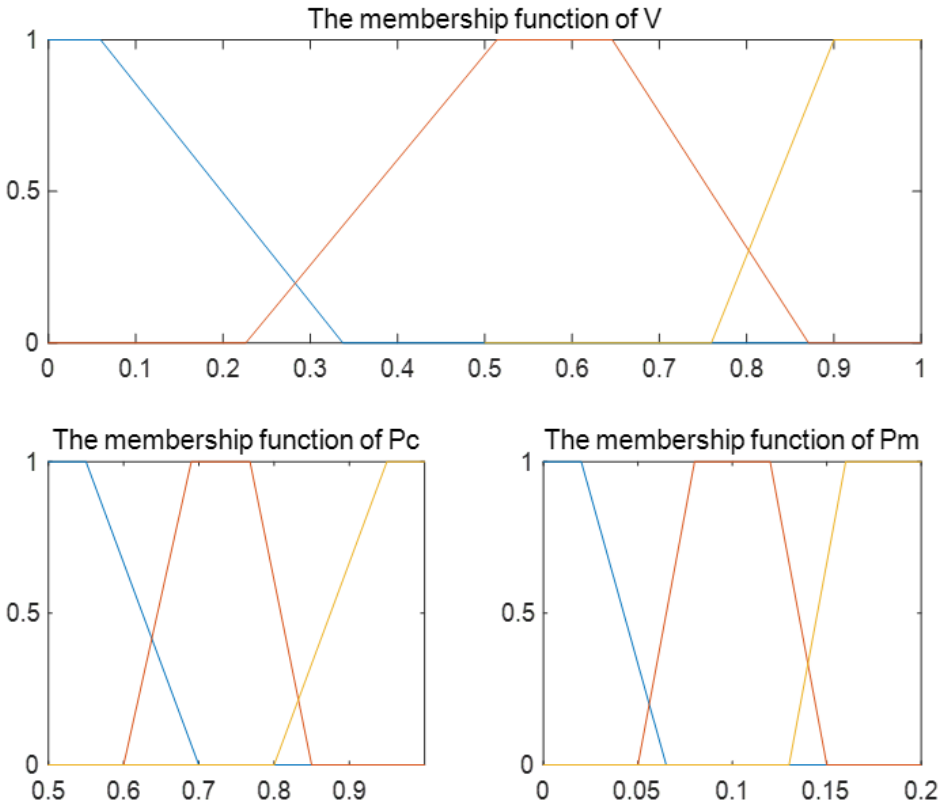


Figure 4. FAGA membership function graph.

When the density of the initial population fitness value is small, that is, the variance is small, and the variance does not change with the increase of the number of iterations, it is necessary to increase P_m to reduce P_c. When the variance of population fitness value is large, then we must increase P_c to reduce P_m so that the population is rapidly concentrated to ensure that the genetic algorithm has a strong optimization ability. We take the mean square deviation F of the fitness value as an input variable of the fuzzy system, and the fuzzy rules are as follows.

Rule i: if F is Q_{i1}, then P_c is W_{i1}, and P_m is W_{i2}, where $i = 1, 2, \cdots, n$ represents the fuzzy rule i, Q_{i1} describes the fuzzy language variable of state F, and W_{i1} describes the fuzzy language variable of the size of P_c and P_m. For the sake of simplicity, each fuzzy language variable takes three fuzzy concepts: large, medium, and small. From the above discussion of the sum value selection law, the following fuzzy rules can be obtained:

(1) If F is small, then P_c is small and P_m is large;

(2) If F is medium, then P_c is medium and P_m is medium;

(3) If F is large, then P_c is large and P_m is small.

The specific flow of the fuzzy adaptive genetic algorithm is as follows:

Step 1: The decision variables x_{ij}^k and y_{ij}^k of the model are binary coded within the interval [0, 1]. Determine the binary encoding length by the encoding formula, divide the optimal path into different individuals, which usually generates a collection of several individuals using a random method, randomly generate an initial population of size N, and set the evolutionary generation.

Step 2: Calculate the cost, time, and carbon emission adaptation of the individual transportation n since the three multi-objective functions are constructed for time, distance, and cost, respectively, the units of variables are not uniform, and the model needs to be dimensionless. We therefore calculate the initial mean square error F of the population.

Step 3: Fuzzify F and use the fuzzy system to find out the values of fuzzy language variables of P_c and P_m.

Step 4: The selection operation is used to select individuals from the parent population retained after the fitness evaluation for inheritance in the next-generation population according to the roulette selection method. The basic principle of the roulette selection method is that the greater the fitness function value of each single individual in the parent population, the greater the chance of being selected for inheritance in the next generation.

Step 5: Based on the crossover probability P_c , some genes in individuals are exchanged with each other in some way to generate new individuals. In this paper, two chromosomes are selected from the parent population separately. Then, the coding strings are obtained according to the transport nodes and transport methods in each line, and the crossover operation is performed on the selected coding strings according to the random matching selection method.

Step 6: Undertake the process of replacing some of the genes coding strings of the trans-transport node and the converted transport mode in an individual with other genes to generate new genes based on the mutant gene probability P_m.

Step 7: P_c and P_m run after defuzzification, update the values of P_c and P_m , update the individuals of the population and judge whether the optimization algorithm satisfies the three multi-objective path optimization models or not; otherwise, return to Step 2.

Finally, we must update the values of P_c and P_m:

$$\begin{cases} P_c(k) = P_c(k-1) + \Delta P_c(k) \\ P_m(k) = P_m(k-1) + \Delta P_m(k) \end{cases} \tag{19}$$

It is stipulated that the value of P_c does not exceed the range [0.1, 0.9], and the value of P_m does not exceed the range [0.03, 0.97].

4.3. Analysing Algorithm Complexity

The FAGA algorithm can effectively improve the search ability of GA by writing fuzzy rules and avoid the population falling into local optimum and causing a premature ending. The algorithm is divided into two parts: the parent population generation and the offspring population optimization to analyze the complexity of the algorithm. Firstly, a population with m (m > l) objective functions and N scales is stratified to generate the first-generation population. The optimal path of the optimization objective is divided into different individuals. The fitness values of the three objectives of all individuals are compared to calculate the initial variance of the population. The second step is to consider the initial variance of the population and fuzzify it. The complexity of the algorithm is O(mN3).

Secondly, the fuzzy goal of the genetic algorithm is achieved. The parent group retained from the fitness evaluation adopts the roulette selection method to select some individuals to be inherited into the next-generation group. The first step is to explore the individuals in the population, store them in the current set of parents, and select the next generation of offspring by crossover and mutation operations. By analogy, we must divide the entire population. The complexity of the algorithm is O(mN2). Finally, the parents and offspring are merged. In the algorithm selection operation, a new parent is generated.

5. Arithmetic Example

This paper uses a transportation network including highways and railways to verify the effectiveness of the FAGA algorithm. Suppose a multimodal transport enterprise undertakes a batch of goods that need to be transported to Germany from Nanchang. This paper chooses Nanchang as the originating city of international container multimodal transport and Berlin as the destination city. It must pass through Zhengzhou, Huaihua, Nanning, Shanghai, Manzhouli, Gwadar, Singapore, Rotterdam, and Warsaw as intermediate node cities, and each node needs to undertake the corresponding loading and unloading tasks. The transportation mode between the two nodes in the network can be divided into single transport methods and multimodal transport methods, as shown in Figure 5, which is the network line used in this paper.

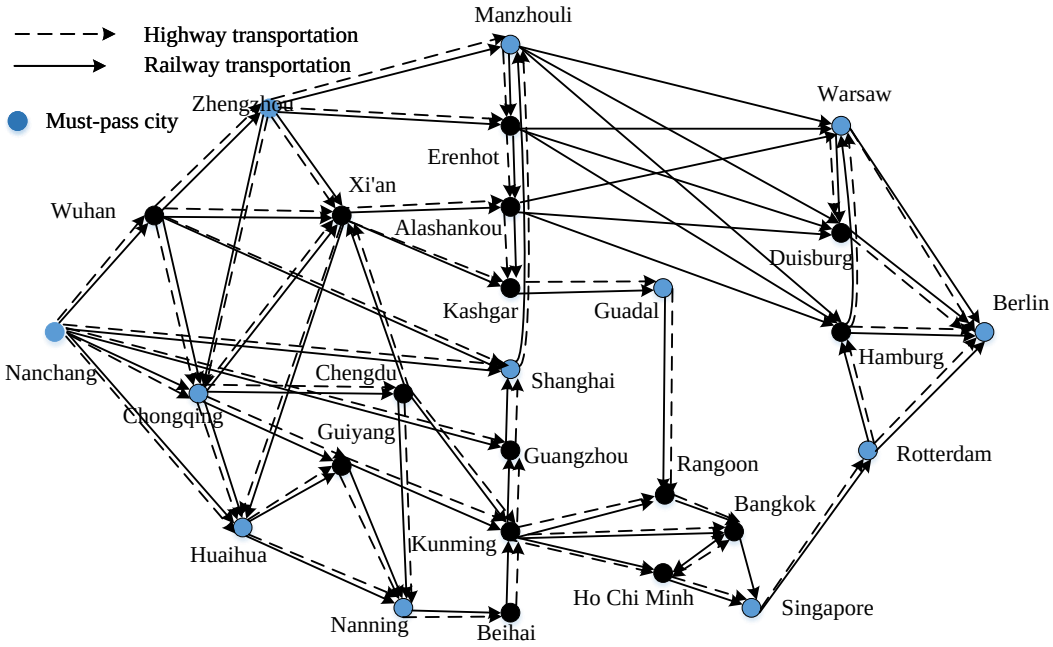


Figure 5. Multimodal transport network from Nanchang to Berlin.

Ten 20-foot containers (with a gross weight of 20 tons) will be transported from Nanchang to Berlin, with 1–2 modes of highway-railway transport available between each of the two connected city nodes. The parameters are set as follows: the maximum number of iterations is 400, the population size is 100, and the freight volume is 20 tons, taking full account of reducing the input cost of logistics enterprises, improving transport efficiency, and reducing carbon emissions. As a result, the route optimization scheme of multimodal transport based on many factors is obtained. The parameters related to switch time and cost and energy consumption of multimodal transport are shown in Table 3. The freight rate, speed and energy consumption of each transport mode are shown in Table 4. The distance between the nodes is shown in Table 5.

Table 3. Switching time, cost, and energy consumption.

Transit Mode	Switching Time (h/box)	Transfer Fee (yuan/box)	Transfer Energy Consumption (kg/box)
Railway-highway	0.5	195	4.025

Table 4. The freight rate, speed and energy consumption of each mode of transportation [41].

Transport Ways	Domestic Part		Foreign Segment		Speed (km/h)	Energy Consumption (kg/box km)
	Base Price 1 (yuan)	Base Price 2 (yuan/box km)	Base Price 1 (yuan)	Base Price 2 (yuan/box km)		
Railway transportation	440	3.185	0	3.211	35	0.072
Highway transportation	0	9.185	0	10.131	43	0.350

5.1. Comparison of Single Mode of Transportation

In order to verify the advantages of the multimodal transport of bulk container goods, this paper compares the results of a single transportation path optimization. In this section, one mode of railway transportation is adopted, and one mode of railway transportation is available between each two urban nodes. This section only considers the mode of railway transportation.

Table 5. Transport mileage of each mode of transportation.

Originating Node	Arrival Node	Railway	Highway
Changsha	Wuhan	358	334
	Huaihua	142	385
	Chongqing	954	901
	Guangzhou	706	670
	Shanghai	1199	1054
Wuhan	Zhengzhou	536	514
	Xi'an	1047	740
	Chongqing	871	866
	Shanghai	1230	820
Chongqing	Huaihua	602	602
	Chengdu	302	300
	Xi'an	728	710
	Guiyang	339	406
Huaihua	Guiyang	314	449
	Nanning	708	675
Zhengzhou	Xi'an	511	479
	Manzhouli	3020	2728
	Erenhot	2137	1173
Xi'an	Alashankou	3045	3037
	Kashgar	3796	3803
Chengdu	Xi'an	842	726
	Kunming	1100	856
	Nanning	1832	1224
Guiyang	Kunming	639	516
	Nanning	865	574
Nanning	Beihai	197	229
Manzhouli	Warsaw	9412	-
	Duisburg	10090	-
	Hamburg	9824	-
Erenhot	Warsaw	8150	-
	Duisburg	8845	-
	Hamburg	8579	-
Alashankou	Warsaw	6278	-
	Duisburg	6955	-
	Hamburg	6689	-
Kashgar	Gwadar	3352	3485
Kunming	Yangon	1542	1638
	Bangkok	1740	1890
	Ho Chi Minh	2056	2164
Yangon	Bangkok	816	923
Ho Chi Minh	Bangkok	928	1027
	Singapore	586	1023
Bangkok	Singapore	2154	2306
Singapore	Rotterdam	820	1364
Warsaw	Berlin	780	1158
Duisburg	Berlin	859	735
Hamburg	Berlin	375	289
Rotterdam	Berlin	778	824

Figure 6 shows the optimization path of the single transportation mode. Nanchang is the starting city, and Berlin is the end city. Wuhan, Zhengzhou, Xi'an, Huaihua, Nanning, Beihai, Kunming, Guangzhou, Shanghai, Manzhouli, Erenhot, Alashankou, Kashgar, Gwadar, Yangon, Bangkok, Singapore, Rotterdam, Hamburg, Warsaw are 20 cities. Table 6 shows the three target values obtained by railway transportation.

5.2. Comparison of Different Algorithms

In this section, a transportation network including highway and railway transportation modes is used to verify the effectiveness of the FAGA algorithm. There are 1–2 modes of transportation in railway transportation and highway transportation between each two urban nodes. In this paper, MATLAB is used to solve the case model. The path optimization effects of general GA, adaptive genetic algorithm (AGA), and the elite genetic algorithm used in [10] are compared with those of this algorithm.

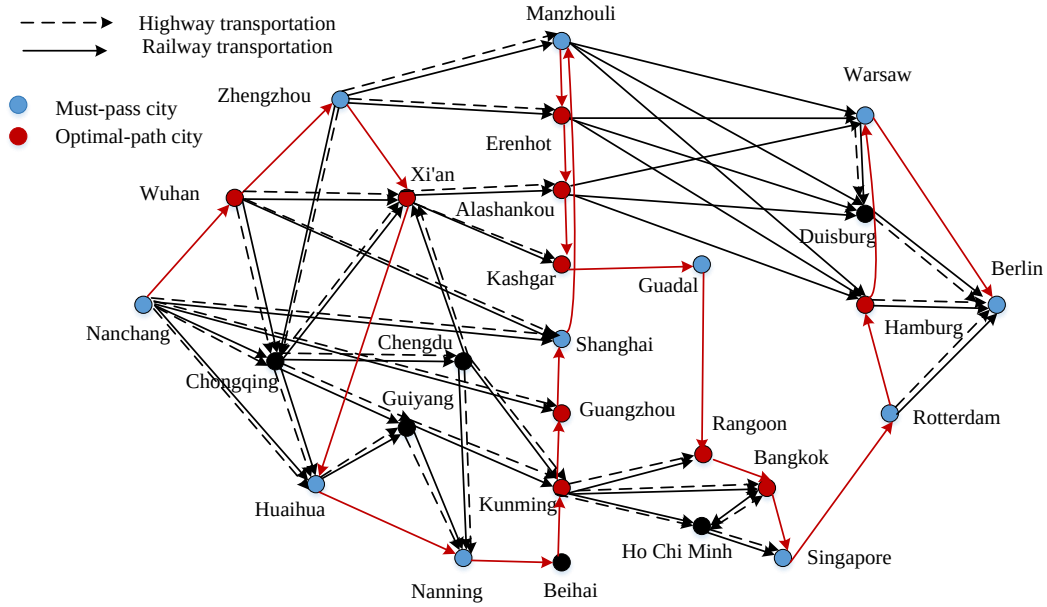


Figure 6. Ordinary GA algorithm to optimize a single transport network from Nanchang to Berlin.

Table 6. Three target values of railway transportation mode.

Goal	Transportation Cost (yuan)	Transport Time (h)	Carbon Emissions (kg)
General Genetic Algorithm (GA)	128,060.65	1138.2	253,281.1

The crossover probability is 0.2, and the mutation probability is 0.8 in GA. Given the maximum genetic algebra gen = 400, the AGA algorithm adaptively adjusts the crossover and mutation parameters according to the fitness of all individuals in each iteration. The FAGA algorithm uses the fuzzy reference system as the mechanism to adaptively select the crossover probability and mutation probability, ensure the diversity of the population, and finally obtain the best route from Nanchang Station to Berlin using the different parameters set above.

Figure 7 shows the optimization path of the ordinary GA algorithm. Nanchang is the starting city, and Berlin is the destination city. It successively passes through 23 cities, including Wuhan, Zhengzhou, Chongqing, Huaihua, Guiyang, Nanning, Beihai, Kunming, Guangzhou, Shanghai, Manzhouli, Erenhot, Alataw Pass, Kashgar, Gwadar, Yangon, Bangkok, Ho Chi Minh, Singapore, Rotterdam, Hamburg, Warsaw, and Duisburg. According to Table 7, a multimodal transportation scheme from Nanchang to Berlin is obtained by switching the mode of transport to highway transport from Nanchang to Wuhan, switching the mode of transport to highway transport from Alashan pass to Kashgar, switching the mode of transport to highway transport from Gwadar to Yangon, switching the mode of transport to highway transport from Ho Chi Minh to Singapore, and finally by highway transport from Duisburg to Berlin. Other modes of transportation are railway transportation. Figures 8–10 show the convergence curves of the iterative process of the three objective values of the ordinary genetic algorithm optimization, from which it can be seen that the cost objective reaches its optimum at around CNY 121,000 in 310 iterations. Similarly, the time objective converges to its optimum at around 660 h in 370 iterations, and the carbon emission objective reaches its optimum at around 92,500 kg in 330 iterations.

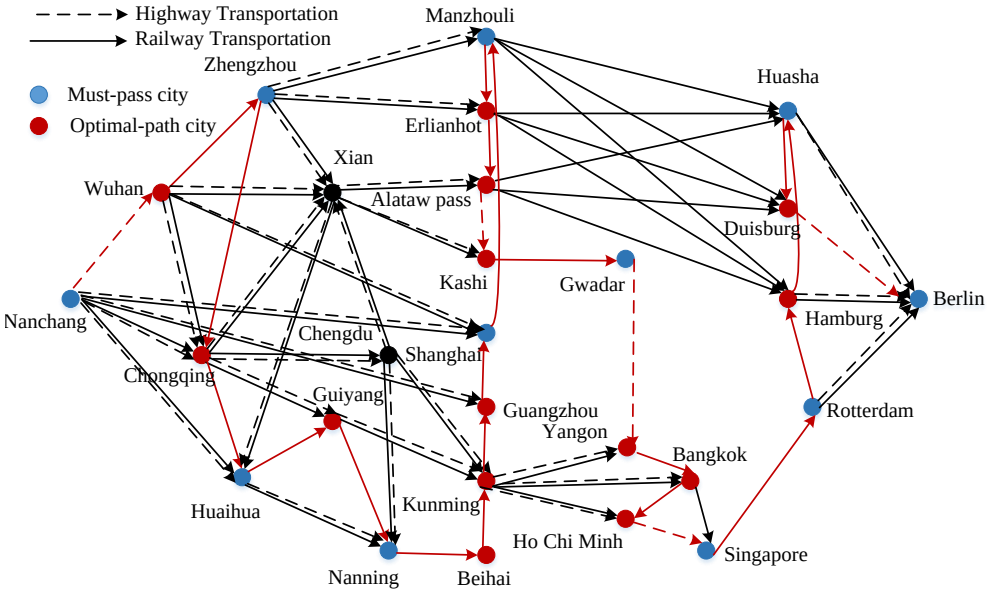


Figure 7. Optimization of Nanchang–Berlin multimodal transport network by general GA algorithm.

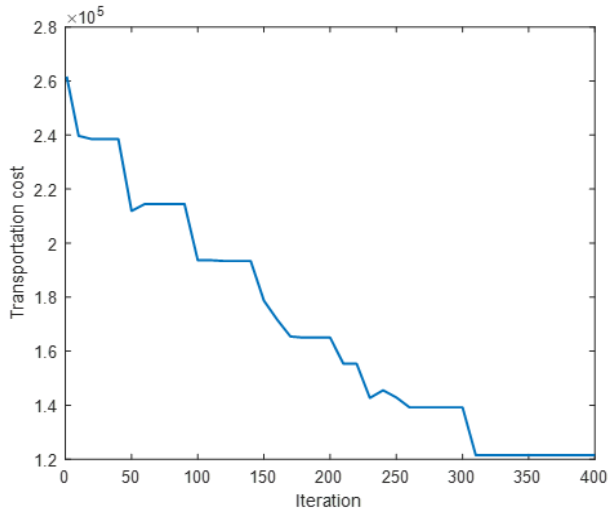


Figure 8. GA algorithm cost target iteration curve.

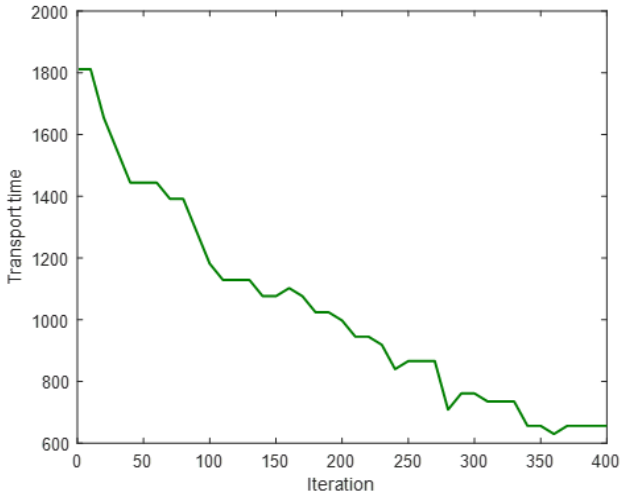


Figure 9. GA algorithm time target iteration curve.

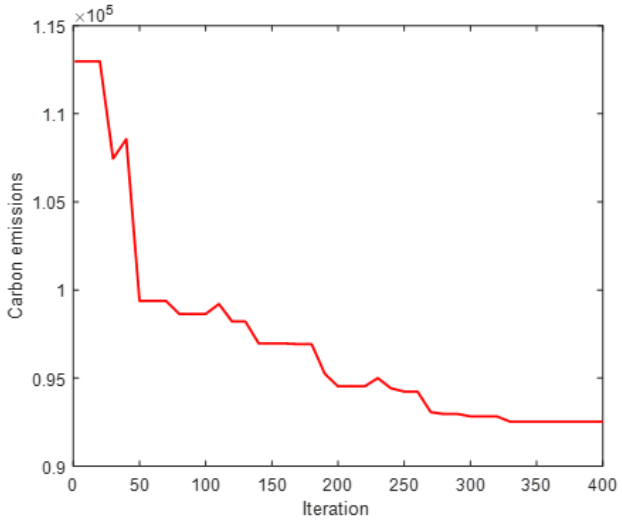


Figure 10. GA algorithm carbon emissions target iteration curve.

Table 7. General GA algorithm to optimize the optimal path of multimodal transport.

City1	Mode of Transportation	City2
Nanchang	Highway	Wuhan
Wuhan	Railway	Zhengzhou
Zhengzhou	Railway	Chongqing
Chongqing	Railway	Huaihua
Huaihua	Railway	Guiyang
Guiyang	Railway	Nanning
Nanning	Railway	Beihai
Beihai	Railway	Kunming
Kunming	Railway	Guangzhou
Guangzhou	Railway	Shanghai
Shanghai	Railway	Manchouli
Manchouli	Railway	Erenhot
Erenhot	Railway	Alataw pass
Alataw Pass	Highway	Kashi
Kashi	Railway	Gwadar
Gwadar	Highway	Yangon
Yangon	Railway	Bangkok
Bangkok	Railway	Ho Chi Minh
Ho Chi Minh	Highway	Singapore
Singapore	Railway	Rotterdam
Rotterdam	Railway	Hamburg
Hamburg	Railway	Warsaw
Warsaw	Railway	Duisburg
Duisburg	Highway	Berlin

Figure 11 shows the path optimized by the AGA algorithm, with Nanchang as the starting city and Berlin as the destination city, through Wuhan, Zhengzhou, Xian, Huaihua, Guiyang, Nanning, Beihai, Kunming, Guangzhou, Shanghai, Manzhouli, Erenhot, Alataw Pass, Kashgar, Gwadar, Yangon, Bangkok, Ho Chi Minh, Singapore, Rotterdam, Hamburg, Warsaw, and Duisburg, a total of 23 cities. According to Table 8, it is concluded that in the scheme of using the AGA algorithm for the multimodal transport from Nanchang to Berlin, the mode of transport is switched to highway transport from Nanchang to Wuhan, from Beihai to Kunming, from Kunming to Guangzhou, from Manzhouli to Erenhot, from Kashgar to Gwadar, from Rotterdam to Hamburg, from Warsaw to Duisburg, and finally highway transportation is used from Duisburg to Berlin. Other modes of transportation are railway transportation. Figures 12–14 show the convergence curves of the iterative process of the three objective values of the AGA genetic algorithm optimization, from which it can be seen that the cost objective reaches its optimum around CNY 114,000 in 300 iterations. Similarly, the time objective converges to its optimum around 660 h in 320 iterations, and the carbon emission objective reaches its optimum around 89,000 kg in 300 iterations.

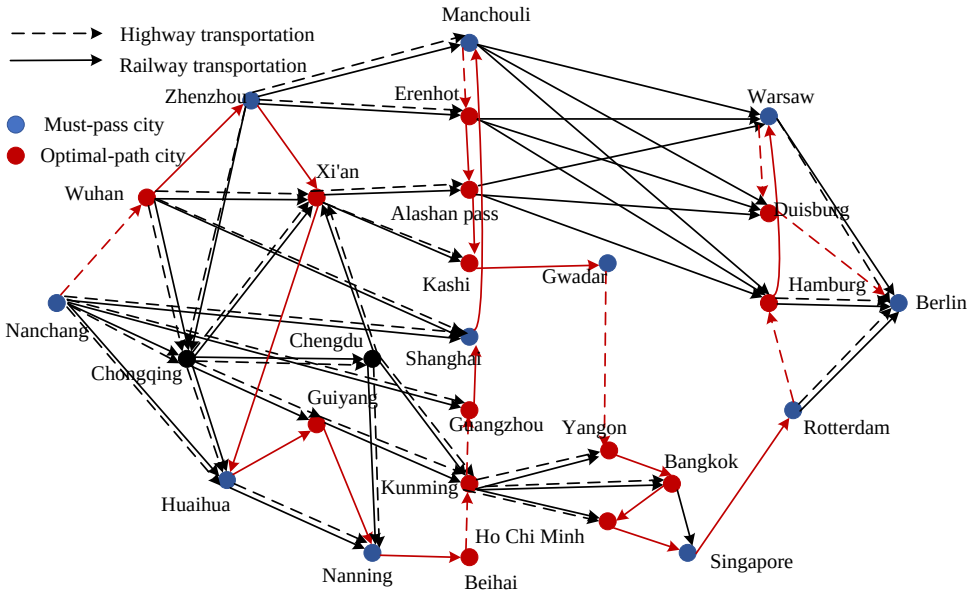


Figure 11. Optimization of Nanchang–Berlin multimodal transport network by AGA algorithm.

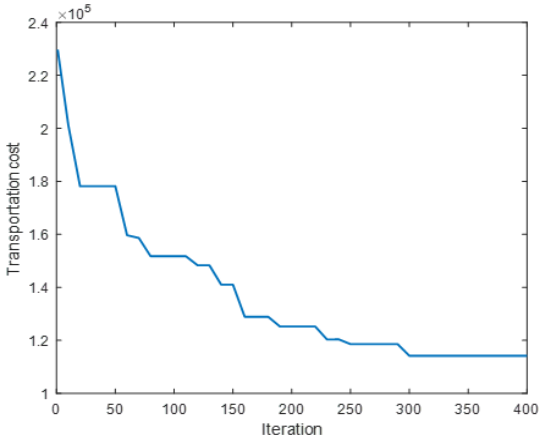


Figure 12. AGA algorithm cost target iteration curve.

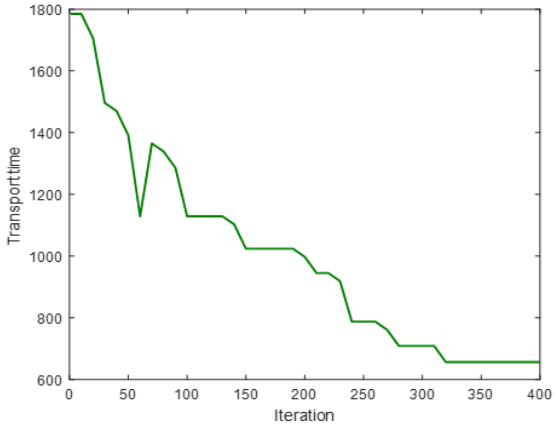


Figure 13. AGA algorithm time target iteration curve.

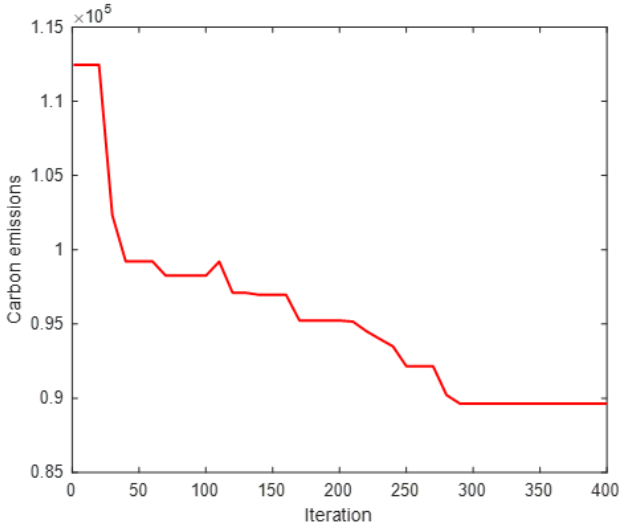


Figure 14. AGA algorithm carbon emissions target iteration curve.

Table 8. Optimizing the optimal path of multimodal transport by the AGA algorithm.

City1	Mode of Transportation	City2
Nanchang	Highway	Wuhan
Wuhan	Railway	Zhengzhou
Zhengzhou	Railway	Chongqing
Chongqing	Railway	Huaihua
Huaihua	Railway	Guiyang
Guiyang	Railway	Nanning
Nanning	Railway	Beihai
Beihai	Highway	Kunming
Kunming	Highway	Guangzhou
Guangzhou	Railway	Shanghai
Shanghai	Railway	Manchouli
Manchouli	Highway	Erenhot
Erenhot	Railway	Alataw pass
Alataw pass	Railway	Kashi
Kashi	Railway	Gwadar
Gwadar	Highway	Yangon
Yangon	Railway	Bangkok
Bangkok	Railway	Ho Chi Minh
Ho Chi Minh	Railway	Singapore
Singapore	Highway	Rotterdam
Rotterdam	Highway	Hamburg
Hamburg	Railway	Warsaw
Warsaw	Highway	Duisburg
Duisburg	Highway	Berlin

Figure 15 shows the optimization path of the algorithm in [10]. Nanchang is the starting city, and Berlin is the destination city. It passes through Wuhan, Zhengzhou, Xi'an, Huaihua, Nanning, Beihai, Kunming, Guangzhou, Shanghai, Manchuria, Erenhot, Alataw Pass, Kashgar, Gwadar, Yangon, Bangkok, Singapore, Rotterdam, Hamburg, Warsaw, and Duisburg. According to Table 9, it is concluded that in the scheme of using the elite genetic algorithm for multimodal transport from Nanchang to Berlin, the conversion of transport mode to highway transport occurs from from Erenhot to Allah Pass, and highway transport is used from Warsaw to Duisburg. Other modes of transport are rail transport. Figures 16–18 shows the convergence curve of the iterative process of cost, time, and carbon emission target values. It can be seen that the cost target is optimal at about CNY 92,394 in 260 iterations. Similarly, the time target is optimal at about 780 h in 325 iterations, and the carbon emissions target is optimal at about 99,105 kg in 250 iterations.

Figure 19 shows the path optimized by the FAGA algorithm, with Nanchang as the starting city and Berlin as the destination city, through Wuhan, Zhengzhou, Chongqing, Huaihua, Nanning, Beihai, Kunming, Guangzhou, Shanghai, Manzhouli, Erenhot, Alataw pass, Kashgar, Gwadar, Yangon, Bangkok, Singapore, Rotterdam, Hamburg, Warsaw, and Duisburg, a total of 21 cities. According to Table 10, it is concluded that in the scheme of using the FAGA algorithm for the multimodal transport from Nanchang to Berlin, the mode of transport is switched to highway transport from Guangzhou to Shanghai, from

Manzhouli to Erenhot, from Erlianhot to Alashankou, and from Singapore to Rotterdam, and finally, highway transportation is used from Warsaw to Duisburg. Other modes of transportation are railway transportation. Figures 20–22 show the convergence curves of the iterative process of the three target values for the optimization of the FAGA algorithm, from which it can be seen that the cost target reaches its optimum at about 260 iterations of CNY 102,000. Similarly, the time target converges to its optimum at about 290 iterations of 630 h, and the carbon emission target reaches its optimum at about 300 iterations of 60,100 kg.

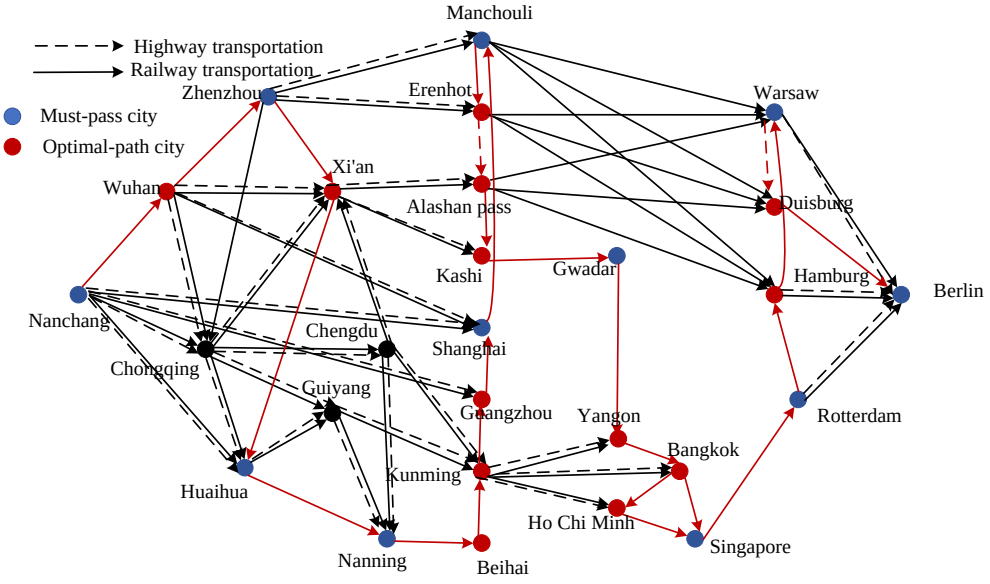


Figure 15. EGA algorithm to optimize multimodal transport network from Nanchang to Berlin.

Table 9. EGA algorithm optimizes the optimal path of multimodal transport.

City1	Mode of Transportation	City2
Nanchang	Railway	Wuhan
Wuhan	Railway	Zhengzhou
Zhengzhou	Railway	Xi'an
Xi'an	Railway	Huaihua
Huaihua	Railway	Nanning
Nanning	Railway	Beihai
Beihai	Railway	Kunming
Kunming	Railway	Guangzhou
Guangzhou	Highway	Shanghai
Shanghai	Railway	Manchouli
Manchouli	Highway	Erenhot
Erenhot	Highway	Alashankou
Alashankou	Railway	Kashi
Kashi	Railway	Gwadar
Gwadar	Railway	Yangon
Yangon	Railway	Bangkok
Bangkok	Railway	Singapore
Singapore	Highway	Rotterdam
Rotterdam	Railway	Hamburg
Hamburg	Railway	Warsaw
Warsaw	Highway	Duisburg
Duisburg	Railway	Berlin

Figure 23 shows the fitness curves of the ordinary GA, AGA, EGA, and FAGA. By comparing the figures, it can be concluded that the fitness function of the FAGA algorithm converges in 6713 iteration over 150 generations, while the GA algorithm converges in 6755 iteration over 325 generations. The EGA algorithm converges in 6730 iterations over 200 generations, and the AGA algorithm converges in 6730 iterations over 160 generations. It can be seen that the FAGA algorithm has a fast convergence speed, good convergence effect, and good optimization quality. In the literature [10], the elite genetic algorithm is designed with the cost as the objective function to solve a given problem. The elite

fragment retention strategy is introduced to enhance the robustness of the algorithm. The elite fragment retention strategy is introduced in the process of gene mutation, and the elite fragment retention is based on the coding of transportation nodes. We apply the model and algorithm in [10] to our example to analyze the optimization results. From the comparison results, our proposed method is superior to the elite genetic algorithm in the improvement of the genetic algorithm. The introduction of the elite genetic algorithm has the advantages of fast convergence speed, a stable optimal solution, and good stability. In order to prevent the optimal solution generated in the evolution process from being destroyed by crossover and mutation, the optimal solution in each generation can be copied to the next generation. The overall convergence speed can be controlled by introducing the proportion of the number of elites. The greater the number, the faster the convergence, but too many elites may cause local convergence of the algorithm, thus obtaining poor results.

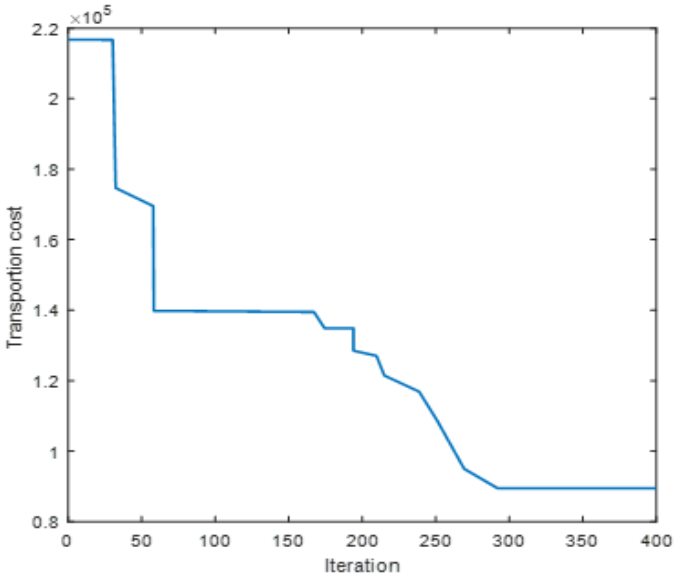


Figure 16. Cost target iteration curve of EGA algorithm.

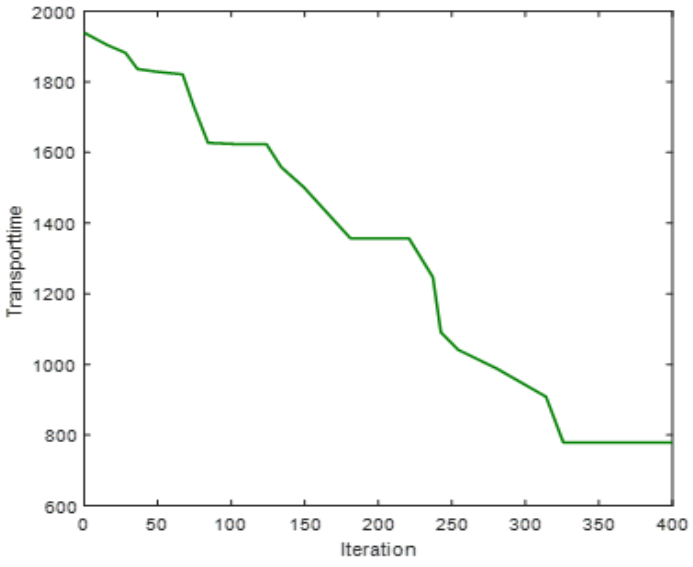


Figure 17. Time target iteration curve of EGA algorithm.

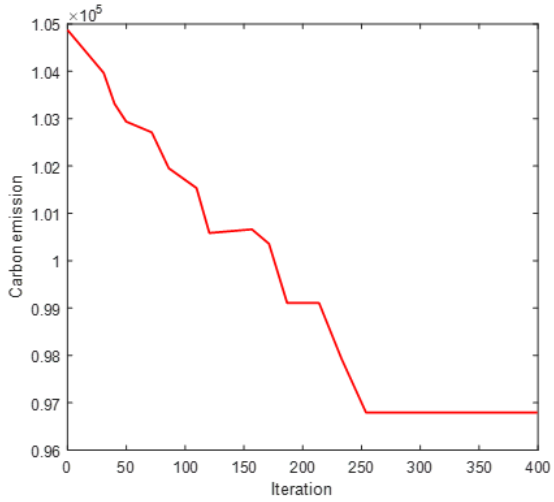


Figure 18. Carbon emissions target iterative curve of EGA algorithm.

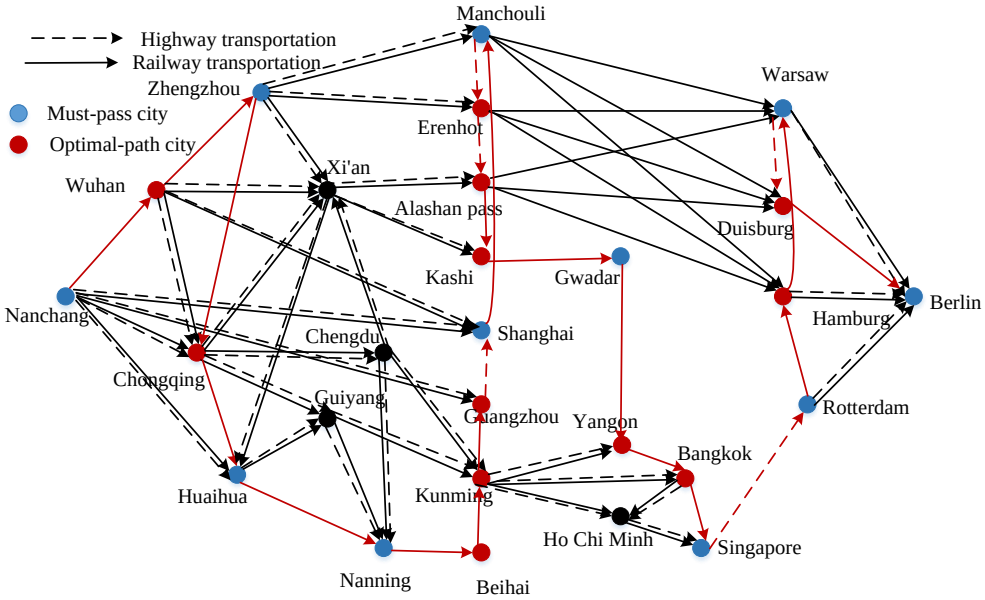


Figure 19. Optimization of Nanchang-Berlin multimodal transport network by FAGA algorithm.

Table 10. Optimizing the optimal path of multimodal transport by the FAGA algorithm.

City1	Mode of Transportation	City2
Nanchang	Railway	Wuhan
Wuhan	Railway	Zhengzhou
Zhengzhou	Railway	Chongqing
Chongqing	Railway	Huaihua
Huaihua	Railway	Nanning
Nanning	Railway	Beihai
Beihai	Railway	Kunming
Kunming	Railway	Guangzhou
Guangzhou	Highway	Shanghai
Shanghai	Railway	Manchouli
Manchouli	Highway	Erenhot
Erenhot	Highway	Alataw pass
Alataw pass	Railway	Kashi
Kashi	Railway	Gwadar
Gwadar	Railway	Yangon
Yangon	Railway	Bangkok
Bangkok	Railway	Singapore
Singapore	Highway	Rotterdam
Rotterdam	Railway	Hamburg
Hamburg	Railway	Warsaw
Warsaw	Highway	Duisburg
Duisburg	Railway	Berlin

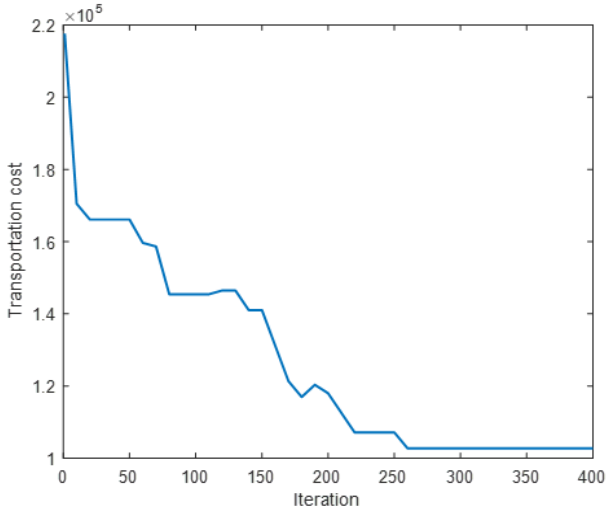


Figure 20. FAGA algorithm cost target iteration curve graph.

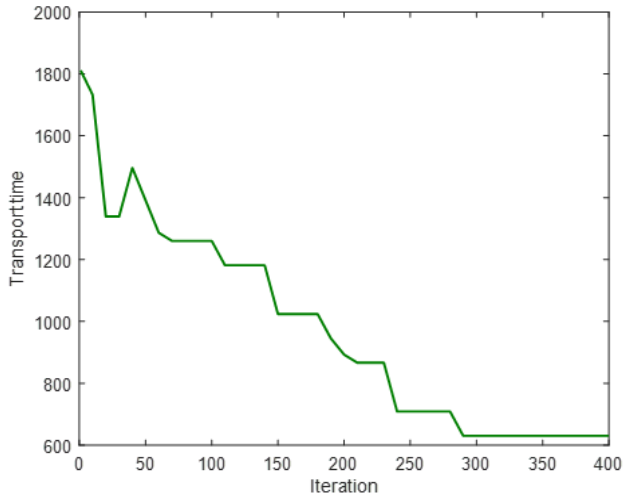


Figure 21. FAGA algorithm time target iteration curve.

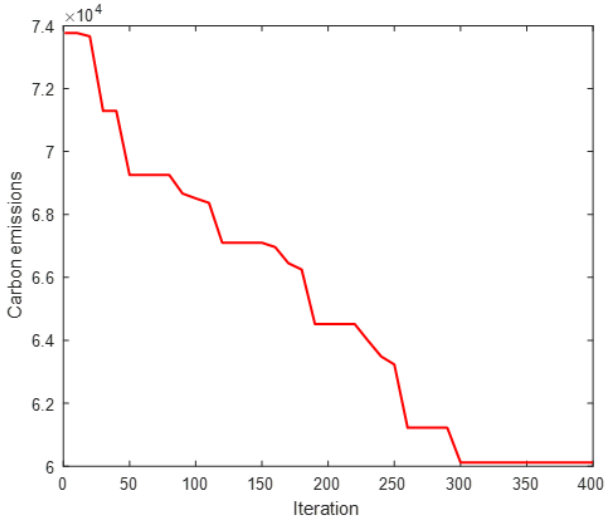


Figure 22. FAGA algorithm carbon emissions target iteration curve graph.

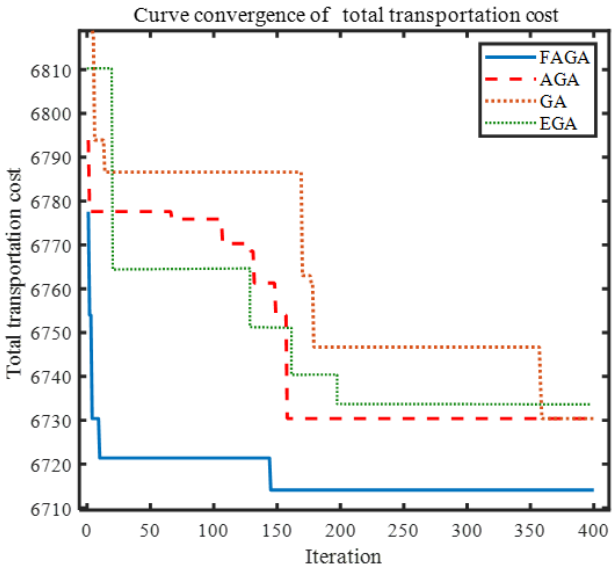


Figure 23. The fitness iterative curve of GA, AGA, EGA, and FAGA.

Therefore, the highway-railway intermodal transportation using the FAGA algorithm saves transportation costs compared with the ordinary GA, AGA and EGA algorithms in this transportation task and can deliver the goods to the destination in advance, reduce carbon emissions, improve customer satisfaction, and truly complete the transportation task at a low cost and quickly. The comparison between the multimodal transport path solved in this section and the single railway transport effect in the previous section can be obtained as shown in Tables 6 and 11. The three target values of multimodal transport have been significantly improved because no matter how far the freight transport distance is, it is completed by several modes of transport together, which can shorten the delivery time of goods, reduce inventory, and improve freight quality. It fundamentally ensures the safe, agile, accurate and timely delivery of goods to the destination.

Table 11. Multi-objective optimization results of GA, AGA, EGA and FAGA algorithms.

Target	Transportation Cost (CNY)	Transport Time (h)	Carbon Emissions (kg)
Genetic algorithm (GA)	121,533.683	660	92,520
Genetic algorithm (AGA)	114,175.316	660	89,640
Elite genetic algorithm (EGA)	92,394.489	780	99,105
Fuzzy genetic algorithm (FAGA)	102,715.564	630	60,120

Finally, this paper compares the improved results of the multiple genetic algorithms involved above and selects the following statistical indicators after 30 runs: fitness function value, success rate, average running time, etc., as shown in Table 12.

Table 12. Statistical indicators of algorithms run 30 times.

Algorithm	Mean Value of Fitness Function	Success Rate	Average Running Time (s)	Optimized Population Standard Deviation
GA	6731.96	0.86	804	336.23
AGA	6729.23	0.75	632	345.59
EGA	6738.11	0.89	554	331.47
FAGA	6713.46	0.92	209	325.18

Through the comparison of data indicators for four improved algorithms based on genetic algorithm, it can be seen that FAGA has achieved good performance based on the genetic algorithm. Because the fuzzy rules used in this study are based on the variance of population fitness value, the three target values of the population will be adjusted in real time during the operation of the algorithm to affect the size of the variance so as

to ensure that the variance and the average fitness between individuals are kept within a reasonable range. Secondly, during the operation of FAGA, the dynamic adjustment of GA parameters or operators is realized, which ensures the reasonable utilization and exploratory optimization of GA performance in the whole GA search process to improve the success rate of operation optimization. Although the complexity of the algorithm is high, the optimization process is effective, and the population quality is high. The automatic adjustment of the algorithm parameters can minimize the resources required for the optimization process, thus reducing the running time and achieving optimization in the fastest time. Based on the EGA algorithm, the optimal individual is directly copied to the next generation without pairing, and the population size is not too large. The success rate, running time, and population standard deviation are significantly improved, and the complexity of the algorithm is low. The adaptive genetic algorithm improves the convergence accuracy of the genetic algorithm by adjusting the parameters of the real-time state of the population, and it does not easily become trapped in the dead cycle phenomenon. The convergence speed is accelerated, so the four indexes are higher than the GA algorithm.

6. Conclusions

This paper starts from the point of view of multimodal transport operators and comprehensively considers transportation cost, transportation time, and carbon emissions. In the case of meeting the carbon emission limit in the carbon emission trading system, the fuzzy adaptive genetic algorithm is used to construct a multi-objective path planning model. Taking the actual transport route from Nanchang to Berlin as an example, an optimal transport route composed of different transport modes under certain constraints is proposed. Finally, the model is optimized by MATLAB software, and the global optimal solution under the three influencing factors of transportation time, transportation cost, and carbon emissions are finally obtained, which provide a reference basis for logistics enterprises to carry out multimodal transport.

The fuzzy adaptive genetic algorithm has lower complexity than the ordinary genetic algorithm and automatically adjusts the selection of crossover probability and variance probability to improve the convergence speed and avoid the problem in which the algorithm can fall into an infinite loop in a local optimum. The decision variables x_{ij}^k and y_{ij}^k of the model are binary coded, and binary-to-decimal conversion is used to calculate the fitness function values of all individuals for convenience in the calculation of the examples. The time complexity does not exceed the squared value of the number of iterations. Therefore, the computational complexity can be effectively reduced. The fuzzy inference system is model-independent. It does not need the mathematical description of the functional relationship between input and output but learns from samples. Secondly, the advantages of the fuzzy reasoning system, such as its parallel structure, simple operation, and small computational requirements, make it very suitable and natural to use the fuzzy reasoning system to describe the crossover probability and mutation probability. At the same time, its reasoning form is easy to understand and can be easily integrated into the existing experience of the multimodal transport route optimization problem model. The fuzzy adaptive genetic algorithm is essentially a stochastic search algorithm based on the biological evolution mechanism, but the computational complexity increases significantly as the problem size increases. In practical application, due to the limitations of actual data and available references, the model fails to find more transportation routes and transportation mode conversions, so the speed of the model and algorithm needs further improvement when the number of nodes increases or the number of genetic maximum iterations increases, which presents certain limitations in practical application.

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